ANALYTICS FRAMEWORK

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INFORMS ANALYTICS FRAMEWORK

The INFORMS Analytics Framework details specific analytics roles, responsibilities and tasks essential for successfully applying analytics to problems, questions, and opportunities enabling impactful business, research, medical, and other decisions. Developed with input from the analytics community, the framework establishes best practices for analytics success.

✅ **Proven Methodology**
The framework mirrors the full analytics lifecycle.

✅ **Strategic Insights**
Use the framework to benchmark your practices and identify actionable steps toward analytics excellence.

✅ **Actionable Guidance from Experts**
Created by INFORMS with analytics leaders and practitioners, with additional input from the analytics community, this framework establishes best practices for analytics

Take the next step toward analytics maturity and organizational success.

Whether you're leading a transformation or refining your analytics strategy, this framework provides the structure and clarity you need to drive measurable impact.

FRAMEWORK DEFINITION:
The INFORMS Analytics Framework was created by experienced Certified Analytics Professionals (CAPs) from diverse industries including government, military, consulting, and academia serving as subject matter experts. Using a detailed job task analysis methodology, they defined core sets of activities related to each domain.

The Seven Domains of Analytics Covered in the INFORMS Analytics Framework

OUTCOME:
The result is a comprehensive framework and structure for all considerations during an analytics approach from initial business problem framing through data collection, model development, deployment, and maintenance. This framework represents best practices for achieving success in the application of analytics to address important business problems and informed decisions. It helps identify areas that need more attention to ensure analytics are effectively applied to key challenges and opportunities.

Domain I: Business Problem (Question) Framing

Analytics methodologies are applied to problems, which are sometimes stated as questions. The **Business Problem (Question) Framing** domain includes activities related to understanding the business problem and relevant questions, as well as evaluating the scope of the problem. The activity may also include the identification of opportunities, challenges, or just an evaluation if a question is well defined. This framing process is essential for ensuring that a clear, actionable problem is defined before seeking solutions. Tasks in this domain include determining, refining and clarifying a business problem statement, identifying stakeholders and aligning the stakeholders with the stated business problem statement, and determining whether the problem is amenable to an analytics solution. With agreement on a business problem statement and appropriate sponsor agreement to move forward with a solution, the remaining domains of the INFORMS Analytics Framework are now applicable.

- Task 1.1 Develop an initial statement of a business problem (question).
- Task 1.2 Identify all stakeholders and their perspectives.
- Task 1.3 Determine if the business problem (question) is amenable to an analytics solution.
- Task 1.4 Refine the initial business problem (question) statement until it is clear and concise.
- Task 1.5 Create an initial business case for a solution to the business problem (question).
- Task 1.6 Obtain sponsor agreement and stakeholder alignment on the business problem (question) statement.

Domain II: Analytics Problem Framing

The **Analytics Problem Framing** domain focuses on translating a clearly defined business problem (question) into a structured analytics problem that can be addressed with analytics techniques applied to data. This process ensures that analytical efforts are aligned with business objectives and that outcomes will be relevant and actionable. A significant activity in this domain is reformulating the business question into an analytics problem, requiring the application of descriptive, predictive or prescriptive analytics. This translation helps define the type of analytics or model needed.

Another activity is identifying the critical drivers or inputs and determining how they relate to desired outputs, forming the basis for hypothesis testing or model design. To ensure relevance and effectiveness, the domain includes defining primary success measures, such as accuracy, revenue lift, or operational efficiency gains. Risk identification and mitigation strategies are also important to anticipate challenges such as data limitations or unintended bias. This domain emphasizes the need for sponsor agreement and stakeholder alignment to ensure buy-in and shared understanding before moving forward with analytics solution development.

By clearly framing the analytics problem, organizations can reduce ambiguity, align technical efforts with strategic goals, and set a solid foundation for model development and implementation.

- Task 2.1 Reformulate the statement of the business problem (question) as an analytics problem statement.
- Task 2.2 Develop a proposed set of drivers/inputs and determine how they relate to outputs.
- Task 2.3 State the set of assumptions related to the analytics problem.
- Task 2.4 Define primary measures of success.
- Task 2.5 Identify baseline performance of the current state.
- Task 2.6 Identify risks and mitigation strategies for an effective potential analytics solution implementation.
- Task 2.7 Obtain sponsor agreement and stakeholder alignment on the Analytics Problem Framing.

Domain III: Data

The **Data** domain encompasses activities related to identifying the data required and available, ensuring its usability for solving analytics problems, and fulfilling documentation and reporting needs. It also includes the effective utilization of data, maintaining its privacy and security, ensuring proper data transport across systems, and adhering to best practices for data governance. Making data usable includes critical actions applied to the data before it can be used such as cleaning, harmonizing, transforming, merging/joining and validating data, as well as data quality evaluation. As a critical asset, security, management, and privacy protocols are vital to support business needs to ensure compliance, ethical standards, and trust in output and outcomes from derived data. This domain also addresses the elements of what data is owned, used, and produced. Understanding the use of data inventory and documentation is incorporated in this domain to ensure repeatable processes and the ability to manage the volume, velocity, variety, and veracity of data.

- Task 3.1 Identify and prioritize data needs.
- Task 3.2 Identify and analyze data sources including data structures.
- Task 3.3 Create a data management plan.
- Task 3.4 Acquire data.
- Task 3.5 Clean, harmonize, transform, merge/join, and validate data.
- Task 3.6 Assess data quality and identify relationships in the data.
- Task 3.7 Document and report data findings (e.g., data quality, impact analysis, results, data management plan.)
- Task 3.8 Validate and update the business and analytics problem statements.

Domain IV: Methodology (Approach) Framing

The **Methodology (Approach) Framing** domain involves selecting the appropriate analytical methods, tools, and technologies to solve the defined analytics problem. This ensures that the chosen approach aligns with the problem type, available data, and resource constraints. The domain includes identifying suitable analytical techniques based on the nature of the problem, as well as evaluating and selecting the best-fit methods considering available data, timeframes. and organizational resources. The domain includes the selection of a compatible technology stack to effectively implement the solution.

Choosing the right methodology is critical to delivering valid, scalable, and actionable insights.

- Task 4.1 Determine available and appropriate methods/techniques for the identified problem.
- Task 4.2 Evaluate and select methods/techniques, given the resources and available data.
- Task 4.3 Understand the technical solution architecture.
- Task 4.4 Evaluate and select the technology stack.

Domain V: Analytics/Model Development

The **Analytics/Model Development** domain covers activities related to designing, building, and refining models, as well as documenting and communicating the results. The domain incorporates designing the model structure, building the appropriate models, running and evaluating model performance to ensure reliability. It also addresses identifying, using, calibrating, and integrating models as well as the documentation of the model and communication of findings. The domain emphasizes documenting and communicating findings, including assumptions and limitations, to maintain transparency.

This domain ensures that models are reliable, well-understood, appropriately tested, and effectively support decision-making.

- Task 5.1 Design the model structure.
- Task 5.2 Build one or more appropriate models.
- Task 5.3 Run, verify, and evaluate the model performance and outputs.
- Task 5.4 Calibrate model and improve data applicability based on client input.
- Task 5.5 If necessary, integrate multiple models.
- Task 5.6 Document and communicate model findings including assumptions, limitations, and constraints.

Domain VI: Deployment

The **Deployment** domain covers activities related to delivering the analytics solution and ensuring it is effectively integrated into the business environment. This involves performing business validation of the solution and a report of the validation findings. It includes securing sponsor agreement and stakeholder alignment to proceed with deployment. The deployment activities support the implementation, migration, and testing of the solution for production, ensuring it functions as intended. Analytics plays a key role in this domain as it involves validation and verification, working in conjunction with technology management, and production data flows. This domain also includes the role of analytics in evaluation of deployment challenges to determine best resolution through data, model, technology, or operational environment conditions. This domain ensures the analytics solution is successfully adopted, integrated, and delivers business value.

- Task 6.1 Perform business validation of the analytics solution.
- Task 6.2 Deliver business validation report with findings.
- Task 6.3 Obtain sponsor agreement and stakeholder alignment on moving forward with deployment.
- Task 6.4 Create requirements for a deployed analytics solution including model, usability, system and business.
- Task 6.5 Actively support the implementation and testing of the production analytics solution/system.
- Task 6.6 Actively support deployment validation and verification, including production data flows.

Domain VII: Analytics Solution Lifecycle Management

Analytics Solution Lifecycle Management domain involves ongoing oversight and calibration to ensure the analytics solution continues to perform effectively and delivers valid results over time that continue to deliver business value. Ongoing training activities are incorporated in this domain to ensure optimal performance management. The domain includes tracking solution performance, covering recalibration and maintenance to keep the solution accurate. Another task validates the business case periodically to confirm continued value, and another task ensures documentation is kept up to date taking into consideration context, environment, and the need for delivering effective end results. Continuous lifecycle management is critical in sustaining the impact and reliability of analytics solutions.

- Task 7.1 Track analytics solution performance.
- Task 7.2 Recalibrate and maintain the analytics solution.
- Task 7.3 Support training activities.
- Task 7.4 Validate the business case for the analytics solution over time.
- Task 7.5 Analyze side effects of the analytics solution over time.
- Task 7.6 Ensure documentation is complete and/or maintained.

CAP CERTIFICATION EXAM BLUEPRINTS and TESTING OBJECTIVES

The INFORMS Analytics Framework details specific analytics roles, responsibilities and tasks essential for successfully applying analytics to challenging problems and opportunities enabling impactful business decisions. It also serves as the basis for the exam blueprints (testing objectives) of all three INFORMS Certified Analytics Professional (CAP®) certification levels. In this section each element of the INFORMS Analytics Framework is broken down into sub-tasks used to define the testing objectives for each of the three levels of the CAP exams.

Using the INFORMS Analytics Framework as a guide, three CAP levels of competency, generally corresponding to different roles within an analytics team, were created corresponding to the CAP certification.

CAP-Essentials (Foundational-level)

CAP-Pro (Mid-level)

CAP-Expert (Advanced level)

The INFORMS Analytics Framework describes seven core domains of knowledge, detailing tasks necessary for professional success within an analytics team. The tasks were then broken down to subtasks, with competence differences across the three role levels elaborated for each task and subtask. The resulting exam blueprints of analytics competencies were validated by a larger sample of practicing analytics professionals.

OUTCOME:

The result is a comprehensive list of subtasks for each task in the INFORMS Analytics Framework. For each subtask, testing objectives are defined for each of the three levels of the Certified Analytics Professional exam. This provides guidance to those interested in achieving the designation of Certified Analytics Professional.

DOMAIN I: BUSINESS PROBLEM (QUESTION) FRAMING

INFORMS Analytics Framework: Analytics methodologies are applied to problems, which are sometimes stated as questions. The **Business Problem (Question) Framing** domain includes activities related to understanding the business problem and relevant questions, as well as evaluating the scope of the problem. The activity may also include the identification of opportunities, challenges, or just an evaluation if a question is well defined. This framing process is essential for ensuring that a clear, actionable problem is defined before seeking solutions. Tasks in this domain include determining, refining and clarifying a business problem statement, identifying stakeholders and aligning the stakeholders with the stated business problem statement, and determining whether the problem is amenable to an analytics solution. With agreement on a business problem statement and appropriate sponsor agreement to move forward with a solution, the remaining domains of the INFORMS Analytics Framework are now applicable.

CAP Testing Objectives:

Task 1.1: Develop an initial statement of a business problem (question).

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.1.1.1	Identify the difference between business statements.	CAP-P.1.1.1	Identify the business problem statement that clearly and completely defines the business problem (question).	CAP-X.1.1.1	Identify and explain questions for obtaining information needed for creating a clear and complete business problem (question) statement.

Task 1.2: Identify all stakeholders and their perspectives.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.1.2.1	Identify the stakeholders.	CAP-P.1.2.1	Identify the primary and secondary stakeholders.	CAP-X.1.2.1	Identify and explain an effective method of conducting stakeholder analysis.
CAP-E.1.2.2	Identify stakeholders and bystanders.	CAP-P.1.2.2	Identify the characteristics of a stakeholder including sponsors, business process owners, information technology, influencers, end users, etc.	CAP-X.1.2.2	Apply responsible, accountable, consulted, and informed criteria to a set of stakeholders.

Task 1.3: Determine if the business problem (question) is amenable to an analytics solution.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E 1.3.1	Identify an example of a simple business problem statement that could be solved with an analytic solution.	CAP-P.1.3.1	Identify an example of an intermediate business problem statement that requires analytics.	CAP-X.1.3.1	Identify and explain an issue that would likely have an impact on the feasibility of an analytics solution.
CAP-E.1.3.2	Not tested at this level.	CAP-P 1.3.2	Identify a reason a problem statement might not be amenable to an analytic solution when framing the business problem.	CAP-X 1.3.2	Identify and explain changes to a problem statement and the reasons that the modified statement would make the problem amenable to an analytic solution.

Task 1.4: Refine the initial business problem (question) statement until it is clear and concise.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E 1.4.1	Identify the part of a business statement that is unclear.	CAP-P.1.4.1	Identify revisions of an unclear business problem statement that provide the most clarity.	CAP-X.1.4.1	Identify revisions necessary to clarify and correct a business problem statement.

Task 1.5: Create an initial business case for a solution to the business problem (question).

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.1.5.1	Identify the definition of business benefits.	CAP-P.1.5.1	Identify an appropriate business benefit or set of benefits.	CAP-X1.5.1	Identify the difference between types of business, societal, and analytical benefits.
CAP-E.1.5.2	Identify the definition of business costs.	CAP-P.1.5.2	Identify the difference between types of business, societal, and project/program costs.	CAP-X.1.5.2	Identify the different types of business costs.
CAP-E 1.5.3	Identify appropriate questions to determine the current state and operating procedures.	CAP-P 1.5.3	Identify inappropriate questions to determine the current state and operating procedures.	CAP-X 1.5.3	Identify methods for determining the current state and operating procedures of business processes that are relevant to the business problem.
CAP-E.1.5.4	Identify an example of a business benefit or cost.	CAP-P.1.5.4	Identify the tradeoffs of business benefits and costs.	CAP-X.1.5.4	Identify how to measure value in a potential solution.
CAP-E 1.5.5	Identify risks to the business of implementing a solution.	CAP-P 1.5.5	Identify the most significant risk to the business of implementing a solution	CAP-X 1.5.5	Identify the most effective approach of risk mitigation to address acceptable risks of implementing an analytics solution.
CAP-E.1.5.6	Identify unintended direct consequences of the potential solution.	CAP-P.1.5.6	Identify indirect consequences of a potential solution to a business problem statement.	CAP-X.1.5.6	Identify options for addressing unintended direct and indirect consequences of a potential analytics solution.

Task 1.6: Obtain sponsor agreement and stakeholder alignment on the business problem (question) statement.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.1.6.1	Identify an example of required sponsor agreement about the business problem, (question) statement.	CAP-P.1.6.1	While obtaining sponsor approval and stakeholder alignment, identify the proposed elements that are in or out of scope of the business problem (question) statement.	CAP-X.1.6.1	Identify how to respond to a concerned party who is either Responsible, Accountable, Supporting, Consulted, or Informed about the business problem (question) statement.

DOMAIN II: ANALYTICS PROBLEM FRAMING

INFORMS Analytics Framework: The **Analytics Problem Framing** domain focuses on translating a clearly defined business problem (question) into a structured analytics problem that can be addressed with analytics techniques applied to data. This process ensures that analytical efforts are aligned with business objectives and that outcomes will be relevant and actionable. A significant activity in this domain is reformulating the business question into an analytics problem, requiring the application of descriptive, predictive or prescriptive analytics. This translation helps define the type of analytics or model needed.

Another activity is identifying the critical drivers or inputs and determining how they relate to desired outputs, forming the basis for hypothesis testing or model design. To ensure relevance and effectiveness, the domain includes defining primary success measures, such as accuracy, revenue lift, or operational efficiency gains. Risk identification and mitigation strategies are also important to anticipate challenges such as data limitations or unintended bias. This domain emphasizes the need for sponsor agreement and stakeholder alignment to ensure buy-in and shared understanding before moving forward with analytics solution development.

By clearly framing the analytics problem, organizations can reduce ambiguity, align technical efforts with strategic goals, and set a solid foundation for model development and implementation.

CAP Testing Objectives:

Task 2.1: Reformulate the statement of the business problem (question) as an analytics problem statement.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.2.1.1	Identify the most appropriate analytics problem statement.	CAP-P.2.1.1	Identify the aspect of a business problem that makes it an analytics problem.	CAP-X.2.1.1	Identify what is missing from a business problem statement that would make it an analytics problem.
CAP-E.2.1.2	Identify if is a statement is a business problem, analytics problem, both or neither.	CAP-P.2.1.2	Identify the most appropriate analytics problem statement.	CAP-X.2.1.2	Identify what information is necessary to translate a business problem statement into one or more analytics problem statement(s) and how the information should be obtained.
CAP-E.2.1.3	Not tested at this level.	CAP-P.2.1.3	Where some analytics research has been performed, assess the research and identify appropriate next steps.	CAP-X.2.1.3	Identify and explain the analytics problem statement that will result in meeting client requirements and achieving client success and project/program goals.

Task 2.2: Develop a proposed set of drivers/inputs and determine how they relate to outputs.

CAP-Essentials (CAP-E)	**Subtask**	**CAP-Pro (CAP-P)**	**Subtask**	**CAP-Expert (CAP-X)**	**Subtask**
CAP-E.2.2.1	Identify analytic inputs and outputs.	CAP-P.2.2.1	Identify why analytics element(s) would be classified as an input, output, both, or neither.	CAP-X.2.2.1	Identify and explain the appropriateness of a proposed set of analytics inputs and outputs.
CAP-E.2.2.2	For a simple business problem statement, identify stated drivers and their relationship to outputs.	CAP-P.2.2.2	Identify how the stated drivers of a moderately complex business problem statement may impact outputs.	CAP-X.2.2.2	Identify and explain the stated drivers or inputs and their relationship to outputs.

Task 2.3: State the set of assumptions related to the analytics problem.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.2.3.1	Identify the explicit assumptions of the business problem that are included in the analytics problem statement.	CAP-P.2.3.1	Identify the implicit assumptions of the business problem that are included in the analytics problem statement.	CAP-X.2.3.1	Identify explicit and implicit assumptions and the implications of those assumptions for appropriate analytical approach(es).
CAP-E.2.3.2	Identify aspects of the business problem that have been simplified.	CAP-P.2.3.2	Identify how to verify if the assumptions for the project/program have been met.	CAP-X.2.3.2	Identify and explain the impacts if the assumptions for a project/program have not been met.
CAP-E.2.3.3	Identify the constraints that should be documented in an analytics problem statement.	CAP-P.2.3.3	Identify how to verify if the business environment limitations have been considered within the context of an analytics problem statement.	CAP-X.2.3.3	Identify examples of the types of constraints that need to be addressed in an analytics problem statement.
CAP-E.2.3.4	Not tested at this level.	CAP-P.2.3.4	Not tested at this level.	CAP-X.2.3.4	Identify limitations to an analytics problem statement with respect to a business problem.

Task 2.4: Define primary measures of success.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.2.4.1	Identify the appropriate primary measures of success from the analytical problem statement that apply to the business statement.	CAP-P.2.4.1	Identify the appropriate analytical problem primary measures of success that apply to the business statement.	CAP-X.2.4.1	Identify the tradeoff and relative importance of the primary measures of success as they relate to business and analytic problem statements.
CAP-E.2.4.2	Identify the primary measures of success of an analytics problem statement.	CAP-P.2.4.2	Identify the process to verify if the primary measures of success have been met.	CAP-X.2.4.2	Identify and explain the process to verify if the primary measures of success have been met.

Task 2.5: Identify baseline performance of the current state.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.2.5.1	Identify how to measure the baseline values of the primary measures of success of the current state.	CAP-P.2.5.1	Identify current baseline performance and how it relates to expected performance as measured by the primary measures of success.	CAP-X.2.5.1	Identify the analytical problem statement that will have the most significant impact on the performance of the current state.

Task 2.6: Identify risks and mitigation strategies for an effective potential analytics solution implementation.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.2.6.1	Identify risks of an analytics solution.	CAP-P.2.6.1	Identify the most significant risks of an analytics solution.	CAP-X.2.6.1	Identify and explain the most effective approach for risk mitigation.
CAP-E.2.6.2	Identify the most likely bias for a predictive model.	CAP-P.2.6.2	Identify the risk of using biased training data for a predictive model to address the business problem.	CAP-X.2.6.2	Identify and explain the most effective approach for mitigating ethical risks.
CAP-E.2.6.3	Not tested at this level.	CAP-P.2.6.3	Not tested at this level.	CAP-X.2.6.3	Identify the risks of using Generative AI as part of the analytics solution.

Task 2.7: Obtain sponsor agreement and stakeholder alignment on the Analytics Problem Framing.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.2.7.1	Not tested at this level.	CAP-P.2.7.1	Identify proposed elements that are in scope for the analytics problem statement while getting sponsor approval and stakeholder alignment.	CAP-X.2.7.1	Identify how to gain alignment on the analytics problem statement from a party who is either Responsible, Accountable, Supporting, Consulted, or Informed.
CAP-E.2.7.2	Not tested at this level.	CAP-P.2.7.2	Not tested at this level.	CAP-X.2.7.2	Evaluate the clarity and completeness in communicating the analytics problem statement including future concerns and sustainability for informed stakeholder alignment.

DOMAIN III: DATA

INFORMS Analytics Framework: The **Data** domain encompasses activities related to identifying the data required and available, ensuring its usability for solving analytics problems, and fulfilling documentation and reporting needs. It also includes the effective utilization of data, maintaining its privacy and security, ensuring proper data transport across systems, and adhering to best practices for data governance. Making data usable includes critical actions applied to the data before it can be used such as cleaning, harmonizing, transforming, merging/joining and validating data, as well as data quality evaluation. As a critical asset, security, management, and privacy protocols are vital to support business needs to ensure compliance, ethical standards, and trust in output and outcomes from derived data. This domain also addresses the elements of what data is owned, used, and produced. Understanding the use of data inventory and documentation is incorporated in this domain to ensure repeatable processes and the ability to manage the volume, velocity, variety, and veracity of data.

CAP Testing Objectives:

Task 3.1: Identify and prioritize data needs.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.1.1	Identify data needs, sources, and acquisition sequence.	CAP-P.3.1.1	Identify an appropriate sequencing and prioritization of data needed, including sources.	CAP-X.3.1.1	Prioritize data needs, sources, and sequence for an analytics project statement.
CAP-E.3.1.2	Identify issues related to sensitive data and restricted usage in collecting and using data.	CAP-P.3.1.2	Identify the costs, benefits and constraints associated with data collection and usage of a data source.	CAP-X.3.1.2	Identify the data requirements, limitations, and priorities when communicating data requirement to stakeholders and providers when conflicts exist.

Task 3.2: Identify and analyze data sources including data structures.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.2.1	Identify limitations and possible constraints of data, given its attributes, context, and metadata.	CAP-P.3.2.1	Identify data limitations and constraints based on data attributes, data context, and metadata, and propose appropriate course of actions.	CAP-X.3.2.1	Based on data attributes, data context, and metadata, identify appropriate courses of action; consider negotiating changes in business processes or different source systems.
CAP-E.3.2.2	Identify characteristics of roles responsible for data governance (data owner, data steward, data custodian).	CAP-P.3.2.2	Identify overlaps and gaps in responsibility, including roles responsible for data governance.	CAP-X.3.2.2	Based on an organization's data strategy, identify and explain how necessary roles should be defined and deconflicted.
CAP-E.3.2.3	Identify the source of the data (system of record, a data warehouse/data lake, manually created, curated spreadsheets, and user input on a screen, and manually recorded data).	CAP-P.3.2.3	Identify what data sources might contain relevant information.	CAP-X.3.2.3	Identify data that is missing from a project and how it might be obtained.
CAP-E.3.2.4	Identify the fundamental strengths and weaknesses of various data architecture.	CAP-P.3.2.4	Identify the appropriate data architecture.	CAP-X.3.2.4	Identify how the analytics technology stack connects and how performance or other needs may shape architecture choices.
CAP-E.3.2.5	Identify the data architecture that should be used.	CAP-P.3.2.5	Identify strengths and weaknesses of data architectures including security, timeliness, and level of effort.	CAP-X.3.2.5	Identify implications of data architectures for analytics projects/programs based on specific attributes.
CAP-E.3.2.6	Identify characteristics of a normalized data set.	CAP-P.3.2.6	Identify basic characteristics of a relational database.	CAP-X.3.2.6	Identify advanced characteristics of a relational database.
CAP-E.3.2.7	Identify which of the 4 Vs (volume, veracity, variety, velocity) are NOT relevant.	CAP-P.3.2.7	Identify which of the 4 Vs (volume, veracity, variety, velocity) are NOT relevant.	CAP-X.3.2.7	Identify the relevance of the 4 Vs (volume, veracity, variety, velocity) in a project.

Task 3.3: Create a data management plan.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.3.1	Identify appropriate ways to handle sensitive data.	CAP-P.3.3.1	Identify the consequences of a poor data strategy, poor data governance for an organization, lack of responsibility for data ownership, and impact on society.	CAP-X.3.3.1	Identify how to work with stakeholders to develop a data strategy and data governance culture.

Task 3.4: Acquire data.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.4.1	Identify techniques that are appropriate for acquiring data.	CAP-P.3.4.1	Identify the techniques appropriate in acquiring the data and identifying data source bias.	CAP-X.3.4.1	Identify the trade offs and risks of different types of data sources and how to address potential issues.
CAP-E.3.4.2	Identify the risks or ethical implications of acquiring unnecessary or unintended data.	CAP-P.3.4.2	Identify which data is likely to have risks or ethical implications if acquired for the project/program.	CAP-X.3.4.2	Identify methods for mitigating or managing data that contains potential risks and ethical implications.
CAP-E.3.4.3	Identify characteristics of lineage, traceability, and version control of data.	CAP-P.3.4.3	Identify the purposes of lineage, traceability, and version control of data.	CAP-X.3.4.3	Identify appropriate methods for establishing lineage, traceability, and version control of data.

Task 3.5: Clean, harmonize, transform, merge/join, and validate data.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.5.1	Identify what it means to clean and transform data, and that merging/joining data across sources may require business rules.	CAP-P.3.5.1	Identify the transformations and merge/joins that may be necessary to solve an analytics problem.	CAP-X.3.5.1	Identify an appropriately designed solution to resolve conflicting needs from data in multiple sources.
CAP-E.3.5.2	Identify common issues in data wrangling, such as missing values, duplicates, redundancy, incorrect/mismatched data types, corrupt data and default data.	CAP-P.3.5.2	Identify how to correct common issues with data.	CAP-X.3.5.2	Identify problems related to cleaning and transforming data in a way to support future modeling.

Task 3.6: Assess data quality and identify relationships in the data.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.6.1	Identify data quality gaps including accuracy, completeness, consistency, timeliness, validity, uniqueness, and outliers.	CAP-P.3.6.1	Identify how to recognize issues with the data based on data quality gaps including missing data, accuracy, completeness, consistency, timeliness, validity, uniqueness, and outliers.	CAP-X.3.6.1	Identify how to correct issues with data that contains quality gaps including missing data, accuracy, completeness, consistency, timeliness, validity, uniqueness, and outliers.
CAP-E.3.6.2	Identify patterns and characteristics of a univariate data set with data profiling outputs	CAP-P.3.6.2	Identify patterns and characteristics of a multivariate data set from data profiling outputs.	CAP-X.3.6.2	Identify patterns and characteristics of data and their implications when dealing with multivariate data and data profiling outputs.
CAP-E 3.6.3	Identify common visualizations and their purpose.	CAP-P.3.6.3	Identify advanced visualizations and their purpose.	CAP-X.3.6.3	Identify and explain the clearest interpretation of a visual representation of data.
CAP-E.3.6.4	Identify the methods to evaluate data quality.	CAP-P.3.6.4	Identify composite features from the data set and how they can improve data quality.	CAP-X.3.6.4	Identify appropriate methods of evaluating data quality issues and how to mitigate them.

Task 3.7: Document and report data findings (e.g., data quality, impact analysis, results, data management plan.)

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.7.1	Identify findings in a report about data sets that may affect analysis.	CAP-P.3.7.1	Identify appropriate elements of a data report.	CAP-X.3.7.1	Identify and explain valid conclusions about a data set's applicability for analytics.

Task 3.8: Validate and update the business and analytics problem statements.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.3.8.1	Identify which finding for univariate data is a quality issue that will result in an update to the analytics problem statement.	CAP-P.3.8.1	Identify appropriate changes to an analytic problem statement based on multiple observed data findings	CAP-X.3.8.1	Identify appropriate changes to an analytic problem statement and the business implications of the changes based on observed data findings.

DOMAIN IV: METHODOLOGY (APPROACH) SELECTION

INFORMS Analytics Framework: The **Methodology (Approach) Framing** domain involves selecting the appropriate analytical methods, tools, and technologies to solve the defined analytics problem. This ensures that the chosen approach aligns with the problem type, available data, and resource constraints. The domain includes identifying suitable analytical techniques based on the nature of the problem, as well as evaluating and selecting the best-fit methods considering available data, timeframes. and organizational resources. The domain includes the selection of a compatible technology stack to effectively implement the solution.

Choosing the right methodology is critical to delivering valid, scalable, and actionable insights.

CAP Testing Objectives:

Task 4.1: Determine available and appropriate methods/techniques for the identified problem.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.4.1.1	Identify the methods/techniques used in each type of analytics (descriptive, diagnostic, predictive, and prescriptive).	CAP-P.4.1.1	Identify the most appropriate methods/techniques that should be used.	CAP-X.4.1.1	Understand and explain the business problem statement to identify the methods/techniques that should used.
CAP-E.4.1.2	Identify the characteristics of descriptive/diagnostic analytics methods/techniques.	CAP-P.4.1.2	Identify which descriptive/diagnostic analytics methods/techniques should be used.	CAP-X.4.1.2	Understand and explain the analytics problem statement to identify which descriptive/diagnostic analytics methods/techniques should be used.
CAP-E.4.1.3	Identify the characteristics of predictive analytics methods/techniques.	CAP-P.4.1.3	Identify which predictive analytics methods/techniques should be used.	CAP-X.4.1.3	Understand and explain the analytics problem statement to identify which predictive analytics methods/techniques should be used.
CAP-E.4.1.4	Identify the characteristics of prescriptive analytics methods/techniques.	CAP-P.4.1.4	Identify which prescriptive analytics methods/techniques should be used.	CAP-X.4.1.4	Understand and explain the analytics problem statement to identify which prescriptive analytics methods/techniques should be used.
CAP-E.4.1.5	Not tested at this level.	CAP-P.4.1.5	Not tested at this level.	CAP-X.4.1.5	Identify how Generative AI could be used to address an analytics problem statement and explain the benefits.

Task 4.2: Evaluate and select methods / techniques, given the resources and available data.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.4.2.1	Identify the technical and business characteristics and uses of specific methodologies (descriptive, diagnostic, predictive, and prescriptive) and techniques.	CAP-P.4.2.1	Identify technical and business benefits of one methodology over another, given limitations on resources and data.	CAP-X.4.2.1	Identify and explain technical costs, business costs, benefits, risks, and implications of choosing different methods/techniques when limitations on resources and data exist.

Task 4.3: Understand the technical solution architecture.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.4.3.1	Identify which component(s) belong in the selection of technical solutions to implement the methodology.	CAP-P.4.3.1	Identify what belongs in the selection of technical solutions to implement the methodology.	CAP-X.4.3.1	Identify and explain an appropriate technical solution needed to employ in a solution set.

Task 4.4: Evaluate and select the technology stack.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.4.4.1	Identify characteristics of the technology stack including databases, analytics software, networking, security, on-premise, cloud, open source vs. proprietary, platforms, etc.	CAP-P.4.4.1	Identify strengths and weaknesses of the technology stack including databases, analytics software, networking, security, on-premise, cloud, open source vs. proprietary, platforms, etc.	CAP-X.4.4.1	Identify strengths, weaknesses, and risks of the technology stack including databases, analytics software, networking, security, on-premise, cloud, open source vs. proprietary, platforms, etc.
CAP-E.4.4.2	Identify the strengths of a spreadsheet analytics model.	CAP-P.4.4.2	Identify the weaknesses of a spreadsheet analytics model.	CAP-X.4.4.2	Identify alternatives to a spreadsheet analytics model and when they are appropriate.

DOMAIN V: ANALYTICS / MODEL DEVELOPMENT

INFORMS Analytics Framework: The **Analytics/Model Development** domain covers activities related to designing, building, and refining models, as well as documenting and communicating the results. The domain incorporates designing the model structure, building the appropriate models, running and evaluating model performance to ensure reliability. It also addresses identifying, using, calibrating, and integrating models as well as the documentation of the model and communication of findings. The domain emphasizes documenting and communicating findings, including assumptions and limitations, to maintain transparency.

This domain ensures that models are reliable, well-understood, appropriately tested, and effectively support decision-making.

CAP Testing Objectives:

Task 5.1: Design the model structure.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.5.1.1	Identify appropriate inputs, outputs, and relationships for simple descriptive/ diagnostic analytics models.	CAP-P.5.1.1	Identify appropriate inputs, outputs, and relationships for descriptive/ diagnostic analytics models.	CAP-X.5.1.1	Identify and explain appropriate inputs, outputs, and relationships for descriptive/ diagnostic analytics models.
CAP-E.5.1.2	Identify appropriate inputs, outputs, and relationships for simple predictive analytics models.	CAP-P.5.1.2	Identify appropriate inputs, outputs, and relationships for predictive analytics models.	CAP-X.5.1.2	Identify and explain appropriate inputs, outputs, and relationships for predictive analytics models.
CAP-E.5.1.3	Identify appropriate inputs, outputs, and relationships for simple prescriptive analytics models.	CAP-P.5.1.3	Identify appropriate inputs, outputs, and relationships for prescriptive analytics models.	CAP-X.5.1.3	Identify and explain appropriate inputs, outputs, and relationships for prescriptive analytics models.

Task 5.2: Build one or more appropriate models.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.5.2.1	Identify the appropriate dependent and independent variables of a predictive model.	CAP-P.5.2.1	Identify the appropriate dependent and independent variables of a predictive model.	CAP-X.5.2.1	Identify what does not belong or is missing from a predictive model and a set of model elements (dependent and independent variables).
CAP-E.5.2.2	Identify the appropriate decision variables, constraints, and objective(s).	CAP-P.5.2.2	Identify the appropriate decision variables, constraints, and objective(s) of a prescriptive model.	CAP-X.5.2.2	Identify what does not belong or is missing from a prescriptive model and a set of model elements (dependent and independent variables).
CAP-E.5.2.3	Identify an error from a list of candidate errors for a predictive model.	CAP-P.5.2.3	Identify an error from a list of candidate errors for a predictive model.	CAP-X.5.2.3	Identify an error and how to rectify the error(s) for a predictive model.
CAP-E.5.2.4	Identify an error from a list of candidate errors for a prescriptive model.	CAP-P.5.2.4	Identify an error from a list of candidate errors for a prescriptive model.	CAP-X.5.2.4	Identify an error and how to rectify the error(s) for a prescriptive model.

Task 5.3: Run, verify, and evaluate the model performance and outputs.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.5.3.1	Identify the correct interpretation of a descriptive/ diagnostic analytics mode output.	CAP-P.5.3.1	Identify the correct interpretation of a descriptive/ diagnostic analytics model output	CAP-X.5.3.1	Identify and explain the correct interpretation with a descriptive/ diagnostic analytics model output.
CAP-E.5.3.2	Identify the appropriate division of training, testing, and validation data for a predictive model.	CAP-P.5.3.2	Identify an appropriate method of cross validation of a predictive analytics model.	CAP-X.5.3.2	Identify the correct ensemble modeling approach for a set of predictive analytics models.
CAP-E.5.3.3	Identify a correct interpretation of the predictive analytics model output.	CAP-P.5.3.3	Identify an appropriate interpretation of the output of a predictive analytics model	CAP-X.5.3.3	Identify an appropriate interpretation of the output from a predictive analytics model.
CAP-E.5.3.4	Identify the correct verification of the solution of a simple prescriptive analytics model output.	CAP-P.5.3.4	Identify the correct verification of the solution of a prescriptive analytics model output.	CAP-X.5.3.4	Identify and explain the correct verification of the solution for a prescriptive analytics model output.
CAP-E.5.3.5	Identify which data or model issue is the cause of an observed bias or unethical outcomes of a simple predictive model	CAP-P.5.3.5	Identify which data or model issue is the cause of an observed bias or unethical outcomes of a moderately complex predictive model.	CAP-X.5.3.5	Identify and explain which data and/or model issue is the cause of an observed biased or unethical outcome of a predictive or prescriptive model.

Task 5.4: Calibrate model and improve data applicability based on client input.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.5.4.1	Identify which data or model issue is the cause of a client concern about the output of a simple predictive analytics model	CAP-P.5.4.1	Identify which data or model issue is the cause of a client concern about the output of a moderately complex predictive analytics model.	CAP-X.5.4.1	Identify and explain the combination of data and model issues that are the cause of a client concern with the output of a predictive analytics model.
CAP-E.5.4.2	Identify which data or model issue is the cause of a client concern about the output of a simple prescriptive analytics model	CAP-P.5.4.2	Identify which data or model issue is the cause of a client concern about the output of a moderately complex prescriptive analytics model.	CAP-X.5.4.2	Identify and explain the combination of data and model issues that are the cause of a client concern with the output of a prescriptive analytics model.

Task 5.5: If necessary, integrate multiple models.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.5.5.1	Identify the business purpose of a specific model from a set of candidate purposes.	CAP-P.5.5.1	Identify a combined set of applicable models from a set of candidate models.	CAP-X.5.5.1	Identify and explain conflicts and synergies between multiple models.

Task 5.6: Document and communicate model findings including assumptions, limitations, and constraints.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.5.6.1	Identify appropriate visualization techniques for communicating the results.	CAP-P.5.6.1	Identify the most appropriate non-technical explanation of the results of a completed model.	CAP-X.5.6.1	Identify and explain appropriate assumptions, limitations and constraints when communicating findings for an audience.

DOMAIN VI: DEPLOYMENT

INFORMS Analytics Framework: The **Deployment** domain covers activities related to delivering the analytics solution and ensuring it is effectively integrated into the business environment. This involves performing business validation of the solution and a report of the validation findings. It includes securing sponsor agreement and stakeholder alignment to proceed with deployment. The deployment activities support the implementation, migration, and testing of the solution for production, ensuring it functions as intended. Analytics plays a key role in this domain as it involves validation and verification, working in conjunction with technology management, and production data flows. This domain also includes the role of analytics in evaluation of deployment challenges to determine best resolution through data, model, technology, or operational environment conditions. This domain ensures the analytics solution is successfully adopted, integrated, and delivers business value.

CAP Testing Objectives:

Task 6.1: Perform business validation of the analytics solution.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.6.1.1	Identify the characteristics of validation when performing a business validation of the analytics solution.	CAP-P.6.1.1	Identify the characteristics of validation when performing a business validation of the analytics solution.	CAP-X.6.1.1	Identify how to perform business validation of the analytics solution and how it should be communicated to the client.
CAP-E.6.1.2	Identify a potential ethical analytics risk.	CAP-P.6.1.2	Identify topics related to ethical analytics that should or should not be included in the business validation report of a business use for a model.	CAP-X.6.1.2	Identify the business use of a model that has the most risk of adverse ethical analytics consequences.

Task 6.2: Deliver business validation report with findings.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.6.2.1	Identify sections which belong in the business validation report.	CAP-P.6.2.1	Identify elements which belong in the business validation report.	CAP-X.6.2.1	Identify a non-technical summary of technical findings that is most appropriate for a non-technical audience.

Task 6.3: Obtain sponsor agreement and stakeholder alignment on moving forward with deployment.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.6.3.1	Not tested at this level.	CAP-P.6.3.1	Not tested at this level.	CAP-X.6.3.1	Identify the characteristics of business impacts of implementing the analytics solution to ensure stakeholder alignment.

Task 6.4: Create requirements for a deployed analytics solution including model, usability, system and business.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.6.4.1	Identify appropriate requirements for the analytics solution to be used in production.	CAP-P.6.4.1	Identify what is missing or does not belong in an outline of the requirements of a production system	CAP-X.6.4.1	Identify and explain the appropriate documentation of a production system for a business user.
CAP-E.6.4.2	Not tested at this level.	CAP-P.6.4.2	Identify characteristics of the model and report documentation so that the analytics solutions can be reused if the business circumstances should change.	CAP-X.6.4.2	Identify characteristics of the model and report documentation necessary so that an analytics solution could be reused if the business circumstances should change.

Task 6.5: Actively support the implementation and testing of the production analytics solution/system.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.6.5.1	Not tested at this level.	CAP-P.6.5.1	Identify the characteristics of deployment testing.	CAP-X.6.5.1	Identify the characteristics of deployment testing and why they are important.
CAP-E.6.5.2	Not tested at this level.	CAP-P.6.5.2	Not tested at this level.	CAP-X.6.5.2	Identify the responsibilities of the analytics professional in the delivery of the production analytics solution or system.

Task 6.6: Actively support deployment validation and verification, including production data flows.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.6.6.1	Not tested at this level.	CAP-P.6.6.1	Identify causes of incorrect data in production systems.	CAP-X.6.6.1	Identify which data sources need to be synchronized during deployment when using multiple data sources.

DOMAIN VII: ANALYTICS SOLUTION LIFECYCLE MANAGEMENT

INFORMS Analytics Framework: The **Analytics Solution Lifecycle Management** domain involves ongoing oversight and calibration to ensure the analytics solution continues to perform effectively and delivers valid results over time that continue to deliver business value. Ongoing training activities are incorporated in this domain to ensure optimal performance management. The domain includes tracking solution performance, covering recalibration and maintenance to keep the solution accurate. Another task validates the business case periodically to confirm continued value, and another task ensures documentation is kept up to date taking into consideration context, environment, and the need for delivering effective end results. Continuous lifecycle management is critical in sustaining the impact and reliability of analytics solutions.

CAP Testing Objectives:

Task 7.1: Track analytics solution performance.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.7.1.1	Identify the metrics that monitor analytics solution performance.	CAP-P.7.1.1	Identify metrics that reflect an acceptable analytics solution performance.	CAP-X.7.1.1	Identify factors that may affect analytics solution performance over time.

Task 7.2: Recalibrate and maintain the analytics solution.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.7.2.1	Not tested at this level.	CAP-P.7.2.1	Identify potential opportunities for recalibration of the analytics solution.	CAP-X.7.2.1	Identify potential opportunities and risks for recalibration of the analytics solution.
CAP-E.7.2.2	Not tested at this level.	CAP-P.7.2.2	Not tested at this level.	CAP-X.7.2.2	Identify and compare alternative solutions when there are changes in the external environment.

Task 7.3: Support training activities.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.7.3.1	Identify the type of training about the analytics solution that is needed for an analytics professional audience.	CAP-P.7.3.1	Identify the type of training that is needed for an IT audience.	CAP-X.7.3.1	Identify the type of training that is needed for business users in different roles.

Task 7.4: Validate the business case for the analytics solution over time.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.7.4.1	Identify what has changed over time for the business case.	CAP-P.7.4.1	Identify which benefit is attributable to the analytics solution over time.	CAP-X.7.4.1	Identify the benefits and costs that are directly attributable to the analytics solution.

Task 7.5: Analyze side effects of the analytics solution over time.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.7.5.1	Identify the importance of reviewing analytic solutions post deployment for unintended consequences.	CAP-P.7.5.1	Identify likely adverse consequences of implementing the analytics solution.	CAP-X.7.5.1	Identify an outcome from a predictive model that exhibits ethical issues but is not readily apparent in the data.

Task 7.6: Ensure documentation is complete and/or maintained.

CAP-Essentials (CAP-E)	Subtask	CAP-Pro (CAP-P)	Subtask	CAP-Expert (CAP-X)	Subtask
CAP-E.7.6.1	Identify the types of documentation needed for various analytics methodologies.	CAP-P.7.6.1	Identify the types of documentation needed for various audiences.	CAP-X.7.6.1	Identify the lifecycle activities that should be documented and monitored.